

Harsh Bordekar

AI Scientist – Physics Models

Physics-Informed ML for Simulation, Surrogate Modelling & Uncertainty Quantification
PhD Candidate, DLR & TU Braunschweig — Aerospace Composites & Structural Simulation

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🐙 github.com/harshbordekar-stack
🏠 Permanent Resident - Germany

PROFILE

Simulation and machine-learning research engineer with 7+ years developing physics-based models for aerospace structures, including 3+ years applying ML and surrogate modelling directly to simulation problems. PhD work builds a probabilistic, physics-based multiscale surrogate framework for CFRP composites – coupling microstructure-informed mesoscale damage models with stochastic spatial defect distributions to propagate failure from ply to structural scale, replacing computationally expensive full-scale FEM with fast, physics-consistent predictions – alongside a certification-aligned failure criterion that the framework is evaluated against. Runs large-scale simulation campaigns on HPC/SLURM using domain-specific solvers (ABAQUS, Nastran/Patran) to generate the high-fidelity data these models depend on, and applies Monte Carlo uncertainty quantification to characterise prediction confidence ahead of design decisions – direct, hands-on experience with the UQ/out-of-distribution estimation this role treats as a first-class concern. Also built a physics-based CNN surrogate for on-demand computational homogenization, exposed through a tool-calling interface, replacing full-scale FEM homogenization runs with fast inference. Comfortable across the applied-ML stack (PyTorch, TensorFlow, scikit-learn) and in Python/Linux/HPC environments day to day, with a track record of verifying and technically reviewing externally produced simulation models against defined acceptance criteria, and of explaining simulation and ML concepts to both engineering and non-technical audiences.

Core Competencies: ML for Simulation & Surrogate Modelling, Physics-Informed Machine Learning, Multiscale / Multi-Fidelity Damage Modelling, Uncertainty Quantification & Out-of-Distribution Estimation, High-Fidelity Simulation Data Generation, FE Solvers (ABAQUS, Nastran/Patran), HPC-Scale Simulation Campaigns (SLURM), Model Verification & Validation, Physics-Based CNN / Surrogate Modelling, Explainable AI, Python

PROFESSIONAL EXPERIENCE

Structural Analysis / Simulation Engineer

DLR & TU Braunschweig

Physics-informed ML for simulation & surrogate modelling (PhD)

Feb 2021 - Present

- (PhD) Developed a probabilistic, physics-based multiscale surrogate framework for leakage-onset prediction in CFRP hydrogen storage tanks, coupling microstructure-informed mesoscale damage models with stochastic spatial defect distributions to propagate failure across ply, laminate, and structural scales – replacing [x] full-scale FEM runs with a physics-consistent surrogate and estimating probability of failure directly from the simulation-generated dataset.

Tools: Python, ABAQUS, FEM, HPC/SLURM, Multiscale Modelling

- (PhD) Derived a certification-aligned leakage failure criterion mapping combined strain/load states to leakage onset, bridging progressive-damage models and system-level tank simulations and providing the physics-based evaluation standard against which the surrogate framework's coverage, accuracy, and failure modes are assessed.

Tools: ABAQUS, Python, FEM post-processing using C++

- Performed Monte Carlo uncertainty quantification over the multiscale simulation outputs to generate leakage-probability-vs-strain relationships, characterising prediction confidence and enabling risk-informed design decisions for next-generation CFRP pressure vessels – direct UQ/OOD experience applied to physics-simulation outputs.

Tools: Python (SciPy, NumPy), Monte Carlo methods, ABAQUS

- Ran large-scale thermal and structural simulation campaigns on HPC infrastructure to evaluate thermal-protection effectiveness for the Callisto reusable-rocket programme's metal upper housing bay, using domain-specific solvers to generate high-fidelity simulation data under production-scale compute constraints.

Tools: ABAQUS, SLURM (HPC), Python, Nastran, Patran

- Built a physics-based CNN surrogate for microstructure-informed computational homogenization, wrapped in an agentic, tool-calling interface that accepts natural-language queries and microstructure images and returns homogenized/effective material properties – replacing full-scale FEM homogenization runs (hours) with on-demand inference (seconds), retrieval/prediction faithfulness validated at [x]% against ground-truth homogenization results.

Tools: Python, Physics-Based CNN, Hierarchical RAG, LLM Agent Framework, Image Processing

- Verified and technically reviewed externally and student-produced simulation models, meshes, and results against defined methods and acceptance criteria, diagnosing failure modes arising from data gaps or modelling limitations and issuing documented sign-off-style findings.

Tools: ABAQUS, Nastran, Python

- Instructed 60+ Master's students in FEM theory and ABAQUS-based CFRP modelling, covering Continuum Damage Mechanics (CDM), Progressive Damage Analysis (PDA), and Cohesive Zone Modelling (CZM) – demonstrated ability to communicate simulation and ML concepts to both technical and non-technical audiences.

Tools: ABAQUS, Automation using Python

Research Intern, Metallic Fatigue & Damage Tolerance

Explainable AI & physics-based fatigue/defect-tolerance assessment

DLR-Braunschweig

Nov 2019 - Jan 2021

- Developed a CT-based, explainable-AI defect-detection framework achieving [x]% classification accuracy ([x] precision / [x] recall) on [x] CT scans of additively manufactured titanium parts, with SHAP, Grad-CAM and LIME outputs used to support engineering sign-off. (*Published: Journal of Intelligent Manufacturing, 2025.*)

Tools: Python (scikit-learn, TensorFlow), SHAP, Grad-CAM, LIME, CT image processing

- Assessed fatigue and defect tolerance of aerospace-grade titanium components using small-defect fatigue models (Murakami, El-Haddad), coupling defect measurements with numerical simulation to translate physics-based fracture-mechanics criteria into allowable-defect acceptance limits.

Tools: Small-defect fatigue models (Murakami, El-Haddad), Python (NumPy/SciPy), FEM (ABAQUS)

Student Research Assistant

Institute of Static and Dynamic

Leibniz University Hannover

Sep 2018 - Oct 2019

- Developed an ML framework for automated detection and quantification of fiber undulations in unidirectional CFRP laminates from image data, generating spatially resolved defect metrics used directly as high-fidelity inputs for probabilistic structural simulation models assessing manufacturing-induced variability.

Tools: MATLAB, Python (scikit-learn, TensorFlow), image processing, NumPy/SciPy

SKILLS SUMMARY

ML for Simulation & Surrogate Modelling	Physics-Informed Machine Learning Multiscale / Multi-Fidelity Damage Modelling Physics-Based CNN Surrogates Computational Homogenization
Uncertainty & Evaluation	Uncertainty Quantification (Monte Carlo) Verification & Validation Physics-Based Evaluation Against Certification Criteria Failure-Mode Diagnosis
Simulation Solvers & HPC	ABAQUS Nastran/Patran Linear & Non-linear FEM HPC / SLURM (Large-Scale Simulation Campaigns) Analytical Stress Methods (Niu, Bruhn)
ML & AI	PyTorch TensorFlow Scikit-learn Explainable AI (SHAP, Grad-CAM, LIME) Hugging Face Transformers (Fine-tuning)
LLMs & Agentic Systems	Retrieval-Augmented Generation (RAG) LangChain LangGraph Model Context Protocol (MCP) Tool-Calling
Programming Languages	Python C++ FORTRAN MATLAB
Data Analysis & Scientific Computing	NumPy Pandas Matplotlib Seaborn Plotly SALib SciPy
APIs & Backend Development	RESTful APIs FastAPI JSON
Version Control & Reproducibility	Git Docker Linux
Application Development	VS Code JupyterLab Google Colab Streamlit Gradio
Languages	English (C1) German (B1)

AWARDS AND EDUCATION

 2nd Prize: Best Paper Award	DLR (German Aerospace Center) 2026
 3rd Prize: Best Paper Award	DLR (German Aerospace Center) 2024
PhD Candidate, Computational Methods in Engineering	DLR & TU Braunschweig, Germany <i>Feb 2021 – [x] (expected)</i>
Dissertation: Physics-Informed, Multiscale Machine Learning for CFRP Composite Structures	
M.Sc. Computational Methods in Engineering	Leibniz University Hannover, Germany <i>April 2018 – Dec 2020</i>
Focus: Numerical Methods, Scientific Machine Learning, Deep Learning	

SELECTED PUBLICATIONS

1. **Bordekar, H.**, et al. “Microstructure-informed Probabilistic Multiscale Modelling of CFRP via Machine-Learning-Derived Stochastic Volume Elements.” *Manuscript in preparation*, 2026.
2. **Bordekar, H.**, et al. “MicroStructAgent: A Physics-Informed Agentic AI Framework for Computational Homogenization and Quality Assurance of Fiber Composite Laminates.” *Manuscript in preparation*, 2026.
3. **Bordekar, H.**, et al. “Microstructure Characterization of Fiber Composite Laminate Using eXplainable Artificial Intelligence.” *Journal of Composite Materials*, vol. 60, no. 1, 2026, pp. 3–25. [DOI](#)
4. **Bordekar, H.**, et al. “eXplainable Artificial Intelligence for Automatic Defect Detection in Additively Manufactured Parts Using CT Scan Analysis.” *Journal of Intelligent Manufacturing*, vol. 36, no. 2, 2025, pp. 957–974. [DOI](#)

REFERENCES

Prof. Dr. Christian Huehne	Abteilungsleiter Funktionsleichtbau. Deutsches Zentrum für Luft- und Raumfahrt. Institut für Systemleichtbau
Prof. Dr. Oliver Voelkerink	Head of Institute., IMA, TU-Braunschweig
Dr. Nicola Cersullo	R&T Project Manager @ Airbus Hamburg